Department of Computer Science

Lab Manual

of

MACHINE LEARNING

MCA521-4

Class Name: IV MCA

Master of Computer Application

2026

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Department Overview

Department of Computer Science of CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape nation's destiny. The training imparted aims to prepare young minds for the challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field.

Vision

The Department of Computer Science endeavours to imbibe the vision of the University "**Excellence and Service"**. The department is committed to this philosophy which pervades every aspect and functioning of the department.

Mission

"To develop IT professionals with ethical and human values". To accomplish our mission, the department encourages students to apply their acquired knowledge and skills towards professional achievements in their career. The department also moulds the students to be socially responsible and ethically sound.

Introduction to the Programme

Master of Computer Applications (MCA) is a post-graduate programme of CHRIST (Deemed to be University) and approved by the All India Council of Technical Education (AICTE). It is a two-year programme spread over six trimesters to bridge the chasm between computer studies and applications, bringing within reach of enthusiastic youngsters an excellent group of experienced and dedicated staff and experts, and an enviable infrastructure. MCA at CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape the nation's destiny. The training programme aims to prepare young minds for challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field. The Department is committed to the motto "Excellence and Service," and this philosophy pervades every aspect of its functioning.

Programme Objective

This programme is conceptualised and designed based on the strong commitment of the department to provide better quality education to the students. The principal objectives of this course are to:

● Heighten technological awareness
● Train future industry specialists
● Encourage effective software development
● Provide research training
● Involve students in innovative research
● Produce graduates with a two-year professional education in Computer Science with technical, professional, and communications skills.

Ethics and Human Values

1. Only proprietary or open-source software would be used for academic teaching and learning purposes.
2. Copying of programs from the internet, friends or from other sources is strictly discouraged since it impairs development of programming skills.
3. Unique Practical (Domain based) exercises ensure that the students don’t get involved in code plagiarism.
4. Projects undertaken by students during the course are done in teams to improve collaborative work and synergy between team members.
5. Projects involve modularization which initiates students to take individual responsibility for common goals.
6. Passion for excellence is promoted among the students, be it in software development or project documentation.
7. Giving due credit to sources during the seminar and research assignment is promoted among the students
8. The course and its design enforce the practice of good referencing technique to improve the sense of integrity.
9. Courses involving group discussions and debates on ethical practices and human values are designed to sensitize the students in dealing with customers and members within the Organization.

Programme Outcomes:
P01: Foundation Knowledge: Apply knowledge of mathematics, programming logic, and coding fundamentals for solution architecture and problem-solving.
P02: Problem Analysis: Identify, review, formulate, and analyse problems primarily focussing on customer requirements using critical thinking frameworks.
P03: Development of Solutions: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
PO4: Modern Tool Usage: Select, adapt, and apply modern computational tools such as the development of algorithms with an understanding of the limitations including human biases.
P05: Individual and Teamwork: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
P06: Project Management and Finance: Use the principles of project management such as scheduling, and work breakdown structure and be conversant with the principles of Finance for profitable project management.
P07: Ethics: Commit to professional ethics in managing software projects with financial aspects.
Learn to use new technologies for cyber security and insulate customers from malware
P08: Life-long learning: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

Programme Educational Objectives(PEO’s)
PEO1: Develop innovative and effective software solutions through research that addresses real-world challenges, grounded in a commitment to continuous learning and adaptability.
PEO2: Advance the knowledge and practices in the field of computer applications through lifelong learning and rigorous research, supporting professional growth and academic excellence.
PEO3: Exhibit ethical leadership, social responsibility to contribute and enhance community well-being by innovating solutions.

MCA521-4 – MACHINE LEARNING

Course Name: MACHINE LEARNING Code: MCA521-4

Total Hours: 75
L + P + T: 3 + 4 + 0
Max Marks: 100
Credits: 4
Course Type: Core Course
Course Category: Theo&Prac

Course Description

Machine Learning is a key area in computer applications that focuses on building models capable of making predictions or informed decisions based on data. A strong understanding of machine learning enables students to analyze and interpret complex datasets, develop predictive models, and extract meaningful insights. This course covers a wide range of machine learning concepts and techniques, including supervised and unsupervised learning. It provides a solid theoretical foundation while emphasizing practical, hands-on experience with algorithms and real-world data. The course equips students with the knowledge and skills required to apply machine learning techniques across various domains using modern tools and methodologies.

Course Objectives

This course enables students to understand the fundamental concepts of Machine Learning, including its core principles, terminology, and methodologies. Students will learn to differentiate between various types of learning algorithms such as supervised, unsupervised, and reinforcement learning, and recognize their appropriate applications.

Course Outcomes

After successful completion of the course, students will be able to:

- Explain the fundamental principles and scope of Machine Learning
- Implement modelling practices in machine learning for better generalisation
- Interpret predictive modelling results and performance of supervised machine learning techniques
- Assess the outcomes and effects of unsupervised machine learning processes
- Deploy robust machine learning models for interdisciplinary applications

Unit-1: INTRODUCTION TO MACHINE LEARNING Teaching Hours: 15 Hours

Introduction to AI – Knowledge Representation – Data – Representation of Data, Data Processing, Data Storage, Data Processing, Types of Data – Exploratory Data Analysis, Visualisations and Interpretation, Outliers

Machine Learning: Definition, Objectives, Components, Features, Applications – Traditional Programming v/s Machine Learning, Types of Machine Learning – Predictive Models, Statistical Inferences – Applications of Machine Learning

Challenges in ML: Insufficient Quantity of Data, Non-representative and Poor-Quality Data, Irrelevant Features, Estimation Estimation, Reducing Human Biases in Model Building, Ethical Use of Technology

Lab Exercises:

1. Data Preprocessing and visualisation
2. Data exploration and inferences

Unit-2: PROCESSES IN MACHINE LEARNING Teaching Hours: 15 Hours

Data Preparation and Model Selection: Effect of Data Preparation: Encoding, Nominal and Ordinal Representation, Feature Transformation and Feature Engineering, Generalisation, Normalisation, Sampling, Using Saved Weights, Model Selection Strategies: Overfitting, Underfitting, Bias-Variance Trade-off, Testing and Validation: Performance measures - Confusion matrix, Precision-Recall Trade off, F1 Score, ROC Curve, AUC, Error Analysis, Model Parameters, Hyperparameters, Hyperparameter Tuning, Cross Validation, Decision Functions: Introduction to Supervised Machine Learning Methods, Decision Functions, Decision Thresholds, Distance Measures, Hyperplanes, Regression & Classification Problems, Loss and Cost Functions, Linear Regression using OLS, Multi-class vs Multi-label, KNN Classification
Lab Exercises:
3. Saving weights of a generalised model.
4. Regression and Classification Evaluation Metrics: Illustration through Linear Regression and KNN Classification

Unit-3: SUPERVISED LEARNING ALGORITHMS Teaching Hours: 15 Hours
Learning through Optimisation: Gradient Descent, Linear Regression, Logistic Regression Computational Learning: Decision Trees, Naive-bayes classification, Support Vector Machines, Ensemble Learners: Learning, Voting, Bagging, Stacking, Boosting, Random Forests, GridSearchCV
Lab Exercises:

5. Regression through Gradient Descent
6. Comparison of Different Classifiers
7. Illustration of Ensemble Learning Methods

Unit-4: UNSUPERVISED LEARNING AND DIMENSIONALITY REDUCTION Teaching Hours: 15 Hours
Introduction: Types of Unsupervised Learning, Challenges in Unsupervised Learning, Concept of Cost Functions in Unsupervised Learning, Clustering Methods: Distance based, Centroid Based, Elbow method to find Optimal Number of Clusters, Silhouette Method, K-Means Clustering, Hierarchical Clustering: Agglomerative and Bottom Up, Applying Supervised Learning after Clustering, Dimensionality Reduction Methods: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)
Lab Exercises:
8. KMeans and Elbow Methods
9. Hierarchical Clustering and Dendrograms
10. Dimensionality Reduction
11. Image Compression using Dimensionality Reduction.

Unit-5 Teaching Hours: 15 Hours
MACHINE LEARNING IN REAL-WORLD
Emerging Techniques: Role of ML in attaining Sustainable Development Goals, Time Series Preprocessing, Blackbox Methods: Neural Networks & Deep Learning, Reinforcement Learning/QLearning, Self Supervised Learning, Quantum Machine Learning, Keras and Tensorflow for

designing Multi Layer Perceptron. Case Studies: CNN, RNN, Advances in Deep Learning, Natural Language Processing, Real Time Systems, Computer Vision, Robotics, Expert Systems, Generative Networks, Large Language Models. Model Deployment: Model Deployment: Connecting Models to User Interface, Deployment of ML Models.

Lab Exercises:

12. Training a Multi-layer perceptron
13. Deploying Trained ML-models

Text and Reference Books

[1] Campesato, O. (2020). Artificial Intelligence, Machine Learning and Deep Learning. Mercury Learning and Information.

[2] Andreas C. Müller and Sarah Guido (2017), "Introduction to Machine Learning with Python A Guide For Data Scientists" O'Reilly book

[3] Ethem Alpaydin (2005), "Introduction to Machine Learning", Prentice Hall of India

[4] Max Kuhn and Kjell Johnson (2018), "Applied Predictive Modelling", Springer

Essential Reading

[1] Stuart Russell and Peter Norvig(2020), "Artificial Intelligence: A Modern Approach" (4th Edition), Pearson

[2] Aurelien Geron(2019), "Hands on Machine Learning with Scikit-learn, Keras and Tensorflow", 2nd Edition, O'Reilly Books

[3] Kevin P. Murphy(2012), "Machine Learning: A Probabilistic Perspective", MIT Press

[4] Hastie, Tibshirani, Friedman(2008), "The Elements of Statistical Learning" (2nd ed), Springer

[5] Ian Goodfellow, Aaron Courville, Yoshua Bengio (2019), "Deep Learning (Adaptive Computation And Machine Learning Series)", MIT Press

Additional Information (Web Resources):

1. Andrew Ng, Deeplearning.ai, Machine Learning Specialisation, offered through Coursera: https://www.deeplearning.ai/courses/machine-learning-specialization/

Evaluation Pattern: CIA - 50% ESE - 50%

LIST OF PROGRAMS

MSCAIML

Sl. no	Title of lab Experiment	Page number	RBT	CO
1	India Air Quality & Crop Yield — EDA Lab Data Preprocessing, Visualisation & Exploration Combined Lab Sheet		L3	CO1, 3
2			L3	CO2, 3
3			L3	CO3
4			L3	CO3
5			L3	CO3
6			L3	CO3, 4
7			L4	CO3
8			L3	CO3
9			L3	CO4
10			L5	CO4

Evaluation Rubrics:

Evaluation Rubrics for each program would be

Submission Guidelines:

- Make a copy of the lab manual template with your <name_reg:no_subject name >,
- Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as lab number.
- Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
- Create a Git Repository in your profile <ML lab-reg no> . Follow a different branch for each lab <Lab 1, Lab 2…>, and push the code to Git. The link should be provided in Google Classroom along with the PDF of the lab manual.
- Upload the PDF to Google Classroom before the deadline.

Lab 1 & 2

Lab Exercise I & II:

Aim

- To investigate whether worsening air quality across Indian states is linked to declining agricultural output by independently choosing, applying, and justifying appropriate data analysis and visualisation techniques on raw, messy, real-world datasets.

Evaluation Rubrics

Submission Guidelines

1. Make a copy of the lab manual template with your <name_reg:no_subject name>
2. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
3. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
4. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 …> and push the code to Git.
5. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
6. Upload the PDF to Google Classroom before the deadline.

Code with Results

TASK 1 :

Import the Necessary Libraries and Loaded the Datasets

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

air_quality = pd.read_csv("city_day.csv")
crop_production = pd.read_csv("crop_production.csv")
```

```
air_quality.head()
```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	NaN
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	NaN	NaN
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	NaN	NaN

Data Inspection and Profiling

```
air_quality.shape
```

```
(29531, 16)
```

```
air_quality.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
 4   NO          25949 non-null  float64
 5   NO2         25946 non-null  float64
 6   NOx         25346 non-null  float64
 7   NH3         19203 non-null  float64
 8   CO          27472 non-null  float64
 9   SO2         25677 non-null  float64
 10  O3          25509 non-null  float64
 11  Benzene     23908 non-null  float64
 12  Toluene     21490 non-null  float64
 13  Xylene      11422 non-null  float64
 14  AQI         24850 non-null  float64
 15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
```

Statistical Summary

```
air_quality.describe()
```

	PM2.5	PM10	NO	NO2	
count	24933.000000	18391.000000	25949.000000	25946.000000	25346.0
mean	67.450578	118.127103	17.574730	28.560659	32.3
std	64.661449	90.605110	22.785846	24.474746	31.6
min	0.040000	0.010000	0.020000	0.010000	0.0
25%	28.820000	56.255000	5.630000	11.750000	12.8
50%	48.570000	95.680000	9.890000	21.690000	23.5
75%	80.590000	149.745000	19.950000	37.620000	40.1
max	949.990000	1000.000000	390.680000	362.210000	467.6

```
air_quality.isnull().sum()
```

```
City              0
Date              0
PM2.5          4598
PM10          11140
NO             3582
NO2            3585
NOx            4185
NH3           10328
CO             2059
SO2            3854
O3             4022
Benzene        5623
Toluene        8041
Xylene        18109
AQI            4681
AQI_Bucket     4681
dtype: int64
```

```
crop_production.head()
```

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
0	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Arecanut	1254.0	2000.0
1	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Other Kharif pulses	2.0	1.0
2	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Rice	102.0	321.0
3	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Banana	176.0	641.0
4	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Cashewnut	720.0	165.0

```
crop_production.shape
```

```
(246091, 7)
```

```
crop_production.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 246091 entries, 0 to 246090
Data columns (total 7 columns):
 #   Column         Non-Null Count   Dtype
---  ------         --------------   -----
 0   State_Name     246091 non-null  object
 1   District_Name  246091 non-null  object
 2   Crop_Year      246091 non-null  int64
 3   Season         246091 non-null  object
 4   Crop           246091 non-null  object
 5   Area           246091 non-null  float64
 6   Production     242361 non-null  float64
dtypes: float64(2), int64(1), object(4)
memory usage: 13.1+ MB
```

```
crop_production.describe()
```

	Crop_Year	Area	Production
count	246091.000000	2.460910e+05	2.423610e+05
mean	2005.643018	1.200282e+04	5.825034e+05
std	4.952164	5.052340e+04	1.706581e+07
min	1997.000000	4.000000e-02	0.000000e+00
25%	2002.000000	8.000000e+01	8.800000e+01
50%	2006.000000	5.820000e+02	7.290000e+02
75%	2010.000000	4.392000e+03	7.023000e+03
max	2015.000000	8.580100e+06	1.250800e+09

```
crop_production.isnull().sum()
```

```
State_Name          0
District_Name       0
Crop_Year           0
Season              0
Crop                0
Area                0
Production       3730
dtype: int64
```

Observation:

The AQI dataset contains several missing pollutant readings, especially PM10 and AQI values. Removing all rows would cause major data loss.

The crop dataset is mostly complete but Production has missing values that must be treated before analysis.

Outliers may also exist because environmental and agricultural measurements can have extreme values.

TASK 2 :

TASK 2: Missing Value Handling

AQI Dataset:

- Numerical pollutant columns contain continuous measurements.
- Median imputation is used because pollution values may contain extreme readings and median is less affected by outliers.

AQI_Bucket:

- It is categorical data, therefore mode value is used.

Crop Dataset:

- Production is numerical agricultural data.
- Median replacement is selected because crop production varies greatly between regions.

Rows are not removed because missing data percentage is significant.

```
#AQL Treatment:
numeric_cols = air_quality.select_dtypes(include=np.number).columns

for col in numeric_cols:
    air_quality[col] = air_quality[col].fillna(air_quality[col].median())
```

```
#AQI_Bucket:
air_quality["AQI_Bucket"] = air_quality["AQI_Bucket"].fillna(
    air_quality["AQI_Bucket"].mode()[0]
)
```

```
#Crop Production Treatment:
crop_production["Production"] = crop_production["Production"].fillna(
    crop_production["Production"].median()
)
```

```
air_quality.isnull().sum()
```

```
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
Xylene        0
AQI           0
AQI_Bucket    0
dtype: int64
```

```
crop_production.isnull().sum()
```

```
State_Name       0
District_Name    0
Crop_Year        0
Season           0
Crop             0
Area             0
Production       0
dtype: int64
```

TASK 3 :

TASK 3: DATA CLEANING AND STANDARDIZATION

To prepare for merging, we must ensure that the geographical names in both datasets are consistent. This includes fixing case sensitivity, removing white spaces, and mapping cities in the AQI dataset to their respective states found in the crop dataset.

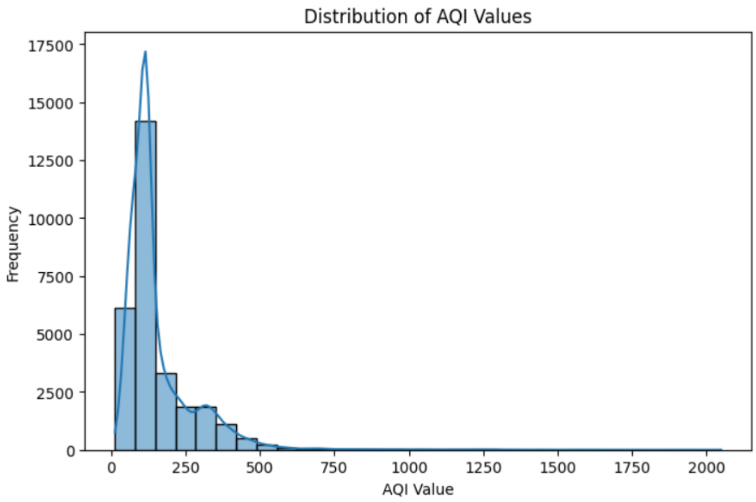
```
for df in [air_quality, crop_production]:
    for col in df.select_dtypes(include=['object']).columns:
        df[col] = df[col].str.strip().str.title()


state_map = {
    'Ahmedabad': 'Gujarat', 'Aizawl': 'Mizoram', 'Amaravati': 'Andhra Pradesh',
    'Amritsar': 'Punjab', 'Bengaluru': 'Karnataka', 'Bhopal': 'Madhya Pradesh',
    'Brajrajnagar': 'Odisha', 'Chandigarh': 'Chandigarh', 'Chennai': 'Tamil Nadu',
    'Coimbatore': 'Tamil Nadu', 'Delhi': 'Delhi', 'Ernakulam': 'Kerala',
    'Gurugram': 'Haryana', 'Guwahati': 'Assam', 'Hyderabad': 'Telangana',
    'Jaipur': 'Rajasthan', 'Jorapokhar': 'Jharkhand', 'Kochi': 'Kerala',
    'Kolkata': 'West Bengal', 'Lucknow': 'Uttar Pradesh', 'Mumbai': 'Maharashtra',
    'Patna': 'Bihar', 'Shillong': 'Meghalaya', 'Talcher': 'Odisha',
    'Thiruvananthapuram': 'Kerala', 'Visakhapatnam': 'Andhra Pradesh'
}

air_quality['State_Name'] = air_quality['City'].map(state_map)

print(f"Records in Crop Data before: {len(crop_production)}")
crop_production.drop_duplicates(inplace=True)
print(f"Records in Crop Data after: {len(crop_production)}")

print(f"Records in AQI Data before: {len(air_quality)}")
air_quality.drop_duplicates(inplace=True)
print(f"Records in AQI Data after: {len(air_quality)}")
```

```
Records in Crop Data before: 246091
Records in Crop Data after: 246091
Records in AQI Data before: 29531
Records in AQI Data after: 29531
```

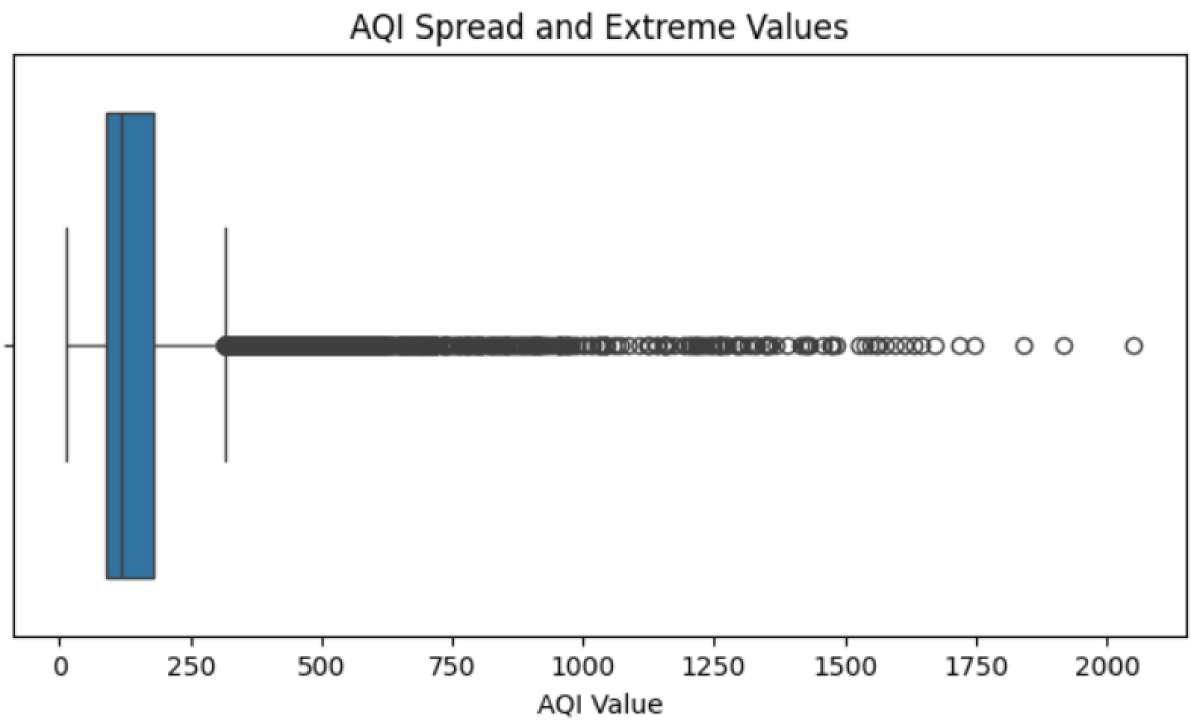

Data Cleaning Summary Standardization: All categorical columns were converted to Title Case and stripped of whitespace to prevent 'Delhi' and ' delhi' from being treated as different entities. Mapping: A new State_Name column was added to the Air Quality dataset. Deduplication: Duplicate records were removed. These steps are critical: without geographical alignment (Mapping) and naming consistency.

TASK 4 :

```
#DISTRIBUTION ANALYSIS
plt.figure(figsize=(8,5))

sns.histplot(
    air_quality["AQI"],
    bins=30,
    kde=True
)

plt.xlabel("AQI Value")
plt.ylabel("Frequency")
plt.title("Distribution of AQI Values")

plt.show()
```



```
plt.figure(figsize=(8,4))

sns.boxplot(
    x=air_quality["AQI"]
)

plt.xlabel("AQI Value")
plt.title("AQI Spread and Extreme Values")

plt.show()
```



Visualization Justification:

Histogram is selected because it shows where most AQI readings are concentrated.

Boxplot is selected because it clearly identifies extreme pollution values.

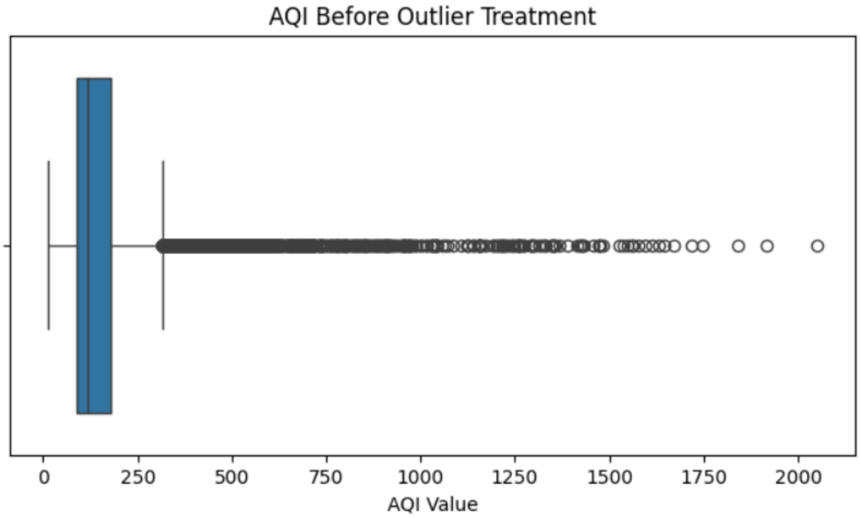
Observations:

1. Most AQI values are concentrated in the lower and moderate range.
2. Some very high AQI readings exist which increase the average AQI.

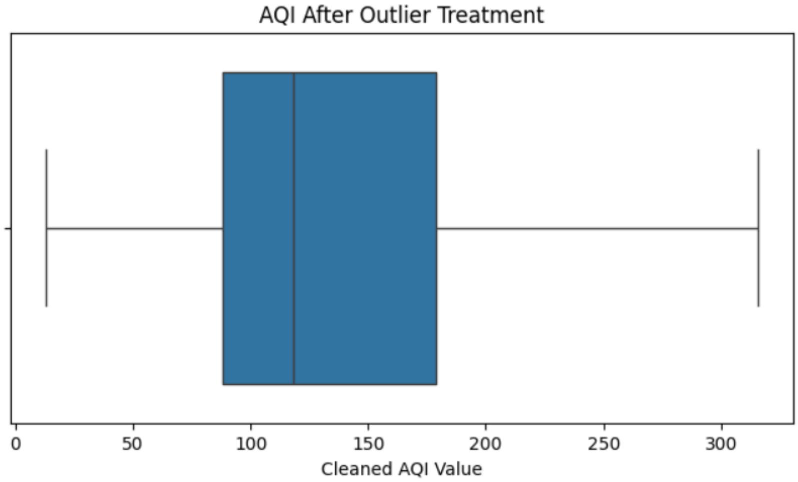
TASK 5 :

```
# Task 5: Outlier Detection and Treatment


# AQI before treatment

plt.figure(figsize=(8,4))

sns.boxplot(
    x=air_quality["AQI"]
)

plt.title("AQI Before Outlier Treatment")
plt.xlabel("AQI Value")

plt.show()



# Calculate IQR

Q1 = air_quality["AQI"].quantile(0.25)

Q3 = air_quality["AQI"].quantile(0.75)


IQR = Q3 - Q1


lower_limit = Q1 - 1.5 * IQR

upper_limit = Q3 + 1.5 * IQR



# Identify extreme values

outliers = air_quality[
    (air_quality["AQI"] < lower_limit) |
    (air_quality["AQI"] > upper_limit)
]


print("Number of extreme AQI values:",
      outliers.shape[0])
```

```
# Treat outliers using capping method

air_quality["AQI_cleaned"] = np.where(
    air_quality["AQI"] > upper_limit,
    upper_limit,
    air_quality["AQI"]
)



# After treatment visualization

plt.figure(figsize=(8,4))

sns.boxplot(
    x=air_quality["AQI_cleaned"]
)

plt.title("AQI After Outlier Treatment")
plt.xlabel("Cleaned AQI Value")

plt.show()
```




```
Number of extreme AQI values: 3192
```



Extreme values were detected using the IQR method.

Instead of deleting records, capping was applied because very polluted days may be real environmental events.

Capping reduces the influence of extreme values while preserving the dataset size for future machine learning.

Lab 2

Data exploration and inferences

TASK 6 :

AQI Trend Over Time

```
# Task 6 : AQI Trend Over Years
# Convert Date column into datetime
air_quality["Date"] = pd.to_datetime(air_quality["Date"])
air_quality["Year"] = air_quality["Date"].dt.year
air_quality[["Date","Year"]].head()
```

	Date	Year
0	2015-01-01	2015
1	2015-01-02	2015
2	2015-01-03	2015
3	2015-01-04	2015
4	2015-01-05	2015

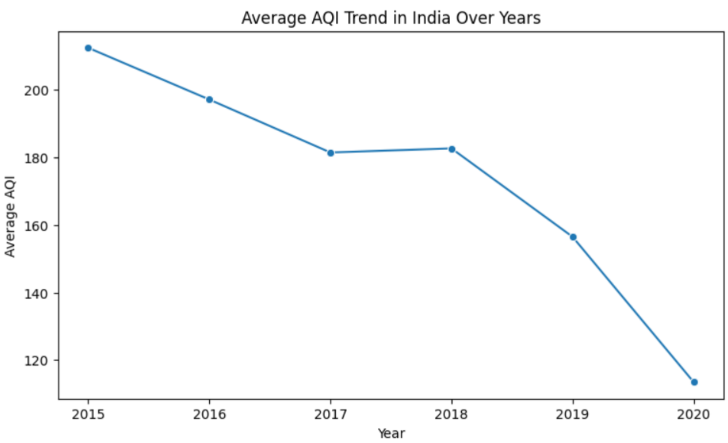
```
yearly_aqi = (
    air_quality
    .groupby("Year")["AQI"]
    .mean()
    .reset_index()
)

yearly_aqi
```

	Year	AQI
0	2015	212.463054
1	2016	197.150019
2	2017	181.472789
3	2018	182.684312
4	2019	156.518173
5	2020	113.520697

```
highest = yearly_aqi[
    yearly_aqi["AQI"] ==
    yearly_aqi["AQI"].max()
]

lowest = yearly_aqi[
    yearly_aqi["AQI"] ==
    yearly_aqi["AQI"].min()
]

print("Most polluted year:")
print(highest)

print("Least polluted year:")
print(lowest)
```

```
Most polluted year:
   Year         AQI
0  2015  212.463054
Least polluted year:
   Year         AQI
5  2020  113.520697
```

```
plt.figure(figsize=(9,5))

sns.lineplot(
    data=yearly_aqi,
    x="Year",
    y="AQI",
    marker="o"
)

plt.title(
    "Average AQI Trend in India Over Years"
)

plt.xlabel("Year")
plt.ylabel("Average AQI")

plt.show()
```



A line plot was selected because it clearly represents changes in AQI over time.

The analysis shows that AQI values changed across years instead of remaining constant. The highest AQI year represents the period with the worst air quality, while the lowest AQI year indicates comparatively cleaner air.

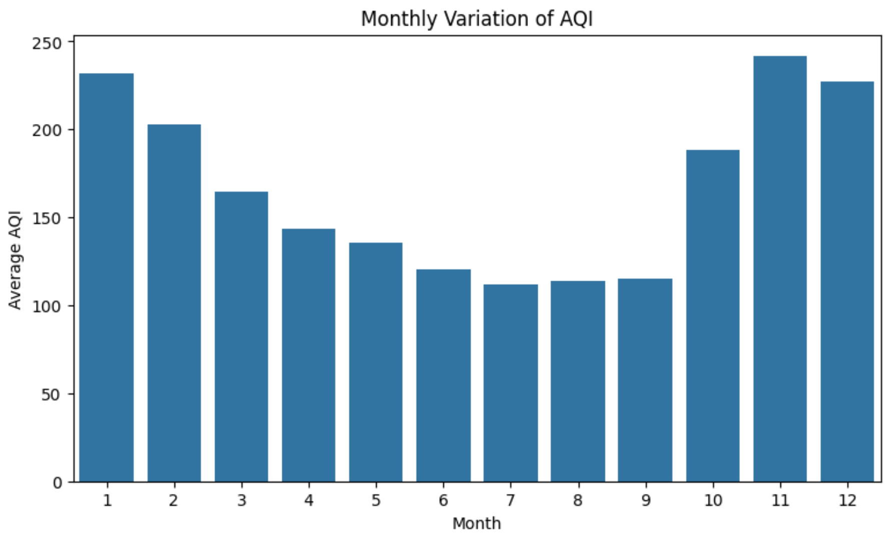
The trend suggests whether pollution control measures after 2018 were associated with improvement or decline in air quality. However, AQI alone cannot prove that policies caused the change because weather and human activities also influence pollution.

TASK 7 :

```
air_quality["Month"] = air_quality["Date"].dt.month
monthly_aqi = (air_quality.groupby("Month")["AQI"].mean().reset_index())
monthly_aqi
```

	Month	AQI
0	1	231.674918
1	2	202.905197
2	3	164.735281
3	4	143.355120
4	5	135.489579
5	6	120.198379
6	7	111.854575
7	8	113.613176
8	9	115.191804
9	10	188.613552
10	11	241.681302
11	12	227.084980

```
plt.figure(figsize=(9,5))

sns.barplot(
    data=monthly_aqi,
    x="Month",
    y="AQI"
)

plt.title(
    "Monthly Variation of AQI"
)

plt.xlabel("Month")
plt.ylabel("Average AQI")

plt.show()
```



TASK 8 :

TASK 8 : Direct merging is not possible because both datasets have different levels of detail.

AQI Dataset:

- Recorded daily at city level

Crop Dataset:

- Recorded yearly at state level

To make them compatible, AQI values are converted into average yearly AQI for each state.

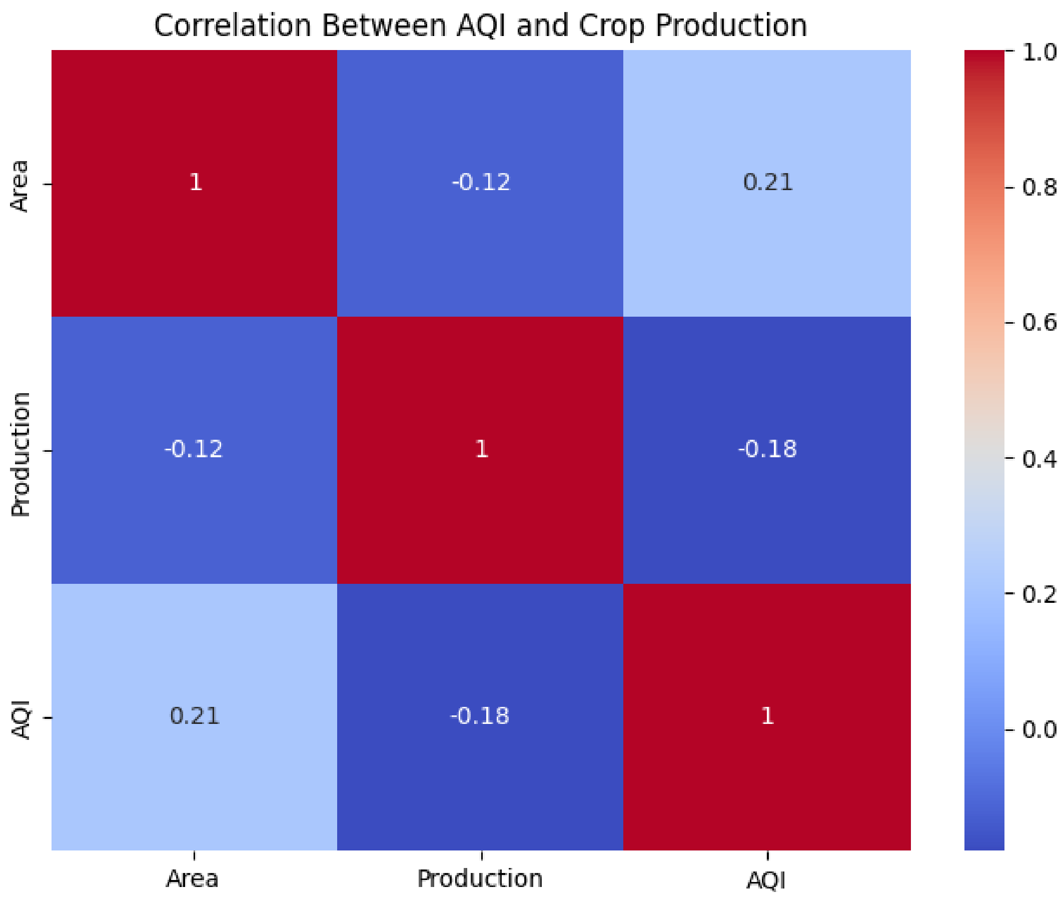
After transformation, both datasets contain State and Year information and can be merged correctly.

```
state_aqi_avg = (air_quality.groupby("State_Name")["AQI"].mean().reset_index())
state_crop_avg = (crop_production.groupby("State_Name")[["Area","Production"]].mean().reset_index())
merged_df = pd.merge(state_crop_avg,state_aqi_avg,on="State_Name",how="inner")
print("Merged Dataset Shape:",merged_df.shape)
merged_df.head()
```

```
Merged Dataset Shape: (20, 4)
```

	State_Name	Area	Production	AQI
0	Andhra Pradesh	13662.842127	1.812006e+06	108.086481
1	Assam	4811.235849	1.444229e+05	140.111111
2	Bihar	6792.270638	1.941738e+04	240.782042
3	Chandigarh	139.133333	7.186124e+02	96.498328
4	Gujarat	18367.001422	6.267679e+04	452.122939

```
corr_matrix = (
    merged_df[
        ["Area","Production","AQI"]
    ]
    .corr()
)


plt.figure(figsize=(8,6))


sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="coolwarm"
)


plt.title(
    "Correlation Between AQI and Crop Production"
)


plt.show()
```




```
Relationship Analysis:

1. AQI vs Production:
The correlation value is -0.18, showing a weak negative relationship.
This means higher pollution levels are slightly associated with lower
crop production, but the effect is not strong.

2. AQI vs Area:
The correlation value is 0.23, showing a weak positive relationship.
States with larger agricultural areas may have slightly higher AQI due
to factors like farming activities, industries, and transportation.

Note:
Correlation does not prove that pollution directly reduces crop yield.
Other factors such as rainfall, irrigation, soil quality, and crop type
also affect agricultural production.
```

TASK 9 :

Policy Brief for the Environment Minister

```
To : State Environment Minster
Subject : Addressing the Air - Quality Agriculture Nexus in india


Our analysis studied India's air quality and agricultural production
data to understand environmental patterns.

The first finding is that air pollution levels have changed over time.
Some years recorded higher AQI levels, showing that continuous pollution
monitoring is important.

The second finding is that air quality changes with seasons. Certain
months show higher pollution levels, meaning farmers and citizens may
experience worse air conditions during specific periods.

The third finding is that crop production and pollution show some
relationships, but agricultural output depends on many factors.

Based on this analysis, one recommended action is to improve pollution
control measures during high AQI months and encourage cleaner farming
practices.

However, this data cannot prove that air pollution alone reduces crop
production. Factors such as rainfall, irrigation facilities, soil
quality and farming methods also strongly influence agricultural
results.

More detailed studies are needed before making final policy decisions.
```

Optional Task :

OPTIONAL TASK B: Relationship Strength Between AQI and Crop Production

The main hypothesis is that states with higher pollution may have lower agricultural production.

To test this relationship, correlation analysis is performed between average AQI and crop production.

Correlation is selected because it measures both:

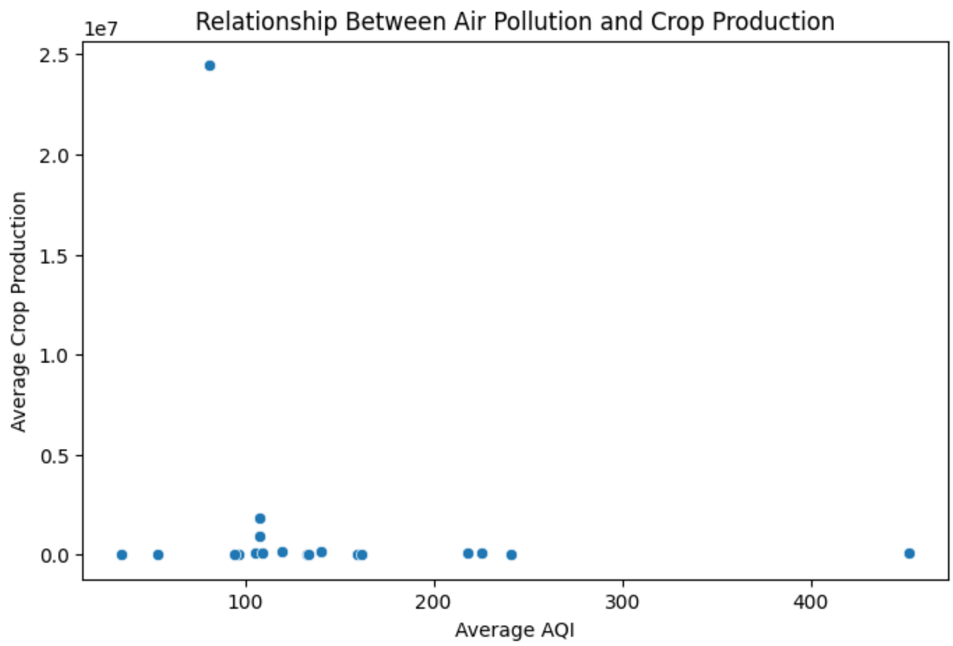
1. Direction of relationship
   - Positive: both variables increase together
   - Negative: one increases while the other decreases
2. Strength of relationship
   - Values closer to 1 or -1 show stronger relationships
   - Values near 0 show weak relationships

However, correlation only shows association and does not prove that pollution causes changes in crop production.

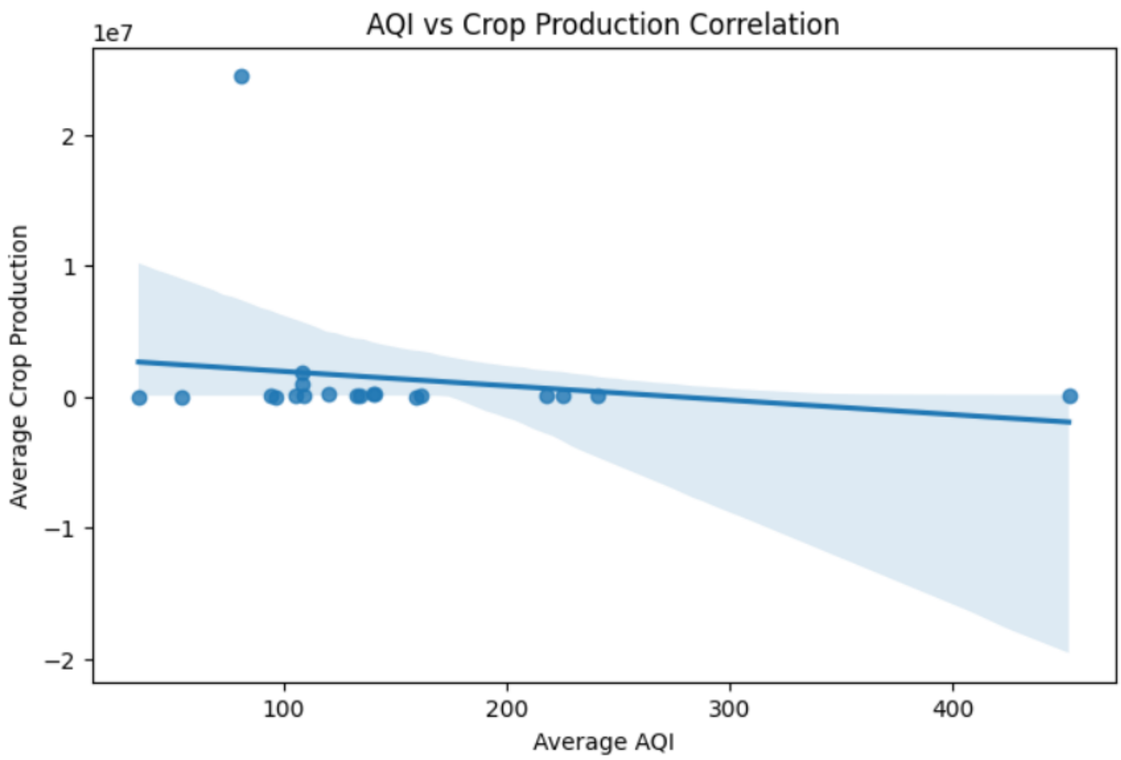
```
aqi_crop_corr = (merged_df["AQI"].corr(merged_df["Production"]))
print("Correlation between AQI and Production:",aqi_crop_corr)
```

```
Correlation between AQI and Production: -0.18036761536447893
```

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=merged_df,x="AQI",y="Production")
plt.title("Relationship Between Air Pollution and Crop Production")
plt.xlabel("Average AQI")
plt.ylabel("Average Crop Production")
plt.show()
```



```
plt.figure(figsize=(8,5))
sns.regplot(
    data=merged_df,
    x="AQI",
    y="Production"
)
plt.title("AQI vs Crop Production Correlation")
plt.xlabel("Average AQI")
plt.ylabel("Average Crop Production")
plt.show()
```



The correlation between average AQI and crop production is -0.18.

The scatter plot and regression line show a weak negative relationship between pollution and agricultural output.

The downward trend suggests that states with higher AQI values may have slightly lower crop production, which provides limited support for the hypothesis that pollution affects agriculture.

However, the points are widely scattered and the correlation value is close to zero, showing that AQI is not the only factor influencing crop production.

This analysis does not prove that pollution causes lower crop yield. Agriculture also depends on rainfall, irrigation facilities, soil quality, crop type and economic conditions.

A more detailed study including these factors is required before confirming a direct impact of air pollution on farming.

Conclusion /inference

This study analyzed the relationship between air quality and agricultural production in India using AQI and crop production datasets through data preprocessing and exploratory data analysis.

During preprocessing, missing values, inconsistent location names, duplicates, and extreme AQI values were identified and handled. A city-to-state mapping was created to combine the air quality and crop datasets for state-level analysis.

The AQI analysis showed that most pollution readings were in the lower to moderate range, but a few extreme values affected the overall average. Time-based analysis showed changes in air quality over the years, while seasonal analysis indicated higher pollution levels during certain periods, especially around the October–December harvest season.

After merging both datasets, a weak negative correlation (-0.18) was found between AQI and crop production. This suggests that higher pollution may be slightly associated with lower agricultural output, but the relationship is not strong.

Overall, air pollution remains an important environmental concern. However, crop production also depends on factors such as rainfall, irrigation, soil quality, and farming methods. Further analysis with more variables is needed to understand the relationship more clearly.